

SOKNEDAL, Norway

WMO: 012530

Lat: 62.953N

Lon: 10.179E

Elev: 981

StdP: 14.18

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.6	(c) 1.6	(d) -6.6	(e) 4.0	(f) -2.3	(g) -2.4	(h) 5.0	(i) 2.4	(j) 18.9	(k) 28.5	(l) 16.4	(m) 27.7	(n) 2.8	(o) 160	(p) 0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	76.7	62.6	72.8	60.2	69.6	58.9	64.3	73.2	62.1	70.2	60.1	67.7	5.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	81.6	68.8	58.5	75.8	66.1	56.6	70.7	63.4	30.0	72.9	28.3	70.6	26.9	68.0	72.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.0	11.8	9.8	-5.7	83.2	6.7	4.4	-10.5	86.4	-14.4	89.0	-18.2	91.5	-23.0	94.7		
			-6.1	67.8	6.6	2.7	-10.9	69.7	-14.7	71.3	-18.4	72.8	-23.2	74.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.5	23.1	25.9	31.0	37.7	46.4	51.8	57.6	55.5	49.6	39.1	29.9	25.4		
	DBStd	14.12	9.73	9.11	7.39	6.13	7.94	5.98	6.23	5.17	5.51	6.11	9.29	9.77		
	HDD50	4493	833	674	590	373	171	48	7	10	75	344	604	764		
	HDD65	9331	1298	1094	1055	820	578	396	244	297	462	804	1054	1229		
	CDD50	654	0	0	0	3	58	102	242	181	63	5	0	0		
	CDD65	17	0	0	0	0	0	1	14	2	0	0	0	0		
	CDH74	265	0	0	0	0	19	25	195	26	0	0	0	0		
	CDH80	51	0	0	0	0	0	2	48	1	0	0	0	0		
	WSAvg	4.0	4.7	4.4	4.4	4.1	4.4	4.0	3.6	3.4	3.7	3.8	3.8	4.1		
	PrecAvg	32.3	2.6	2.4	2.3	1.7	2.0	3.0	3.4	3.4	3.1	2.8	2.7	2.8		
(15) Precipitation	PrecMax	40.8	6.8	4.4	5.5	5.6	3.3	5.9	5.3	8.0	7.4	6.5	7.9	6.5		
	PrecMin	24.1	0.0	0.0	0.7	0.6	0.8	1.1	1.7	0.9	1.1	0.9	0.1	0.6		
	PrecStd	4.5	1.8	1.2	1.5	1.1	0.7	1.2	1.2	1.8	1.8	1.2	2.4	1.5		
	PrecStd	4.5	1.8	1.2	1.5	1.1	0.7	1.2	1.2	1.8	1.8	1.2	2.4	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.1	45.4	49.5	64.6	76.9	77.5	85.9	77.4	69.3	60.7	49.1	43.2		
		MCWB	38.8	39.6	41.6	50.9	59.3	61.2	66.4	63.4	57.9	54.6	45.2	39.1		
	2%	DB	39.6	42.0	45.6	58.1	71.9	71.6	79.7	73.0	65.6	54.9	46.2	41.1		
		MCWB	36.5	37.3	39.8	46.2	56.8	58.9	65.3	60.8	57.0	49.0	42.5	37.2		
	5%	DB	37.7	39.6	43.2	52.1	67.9	67.4	74.5	69.5	62.5	51.5	43.4	39.2		
		MCWB	35.1	35.5	38.2	42.7	55.1	57.1	62.2	59.6	55.7	46.9	40.2	35.8		
	10%	DB	35.6	37.4	40.8	47.9	62.0	63.7	70.4	66.1	59.2	48.0	40.9	37.1		
		MCWB	33.5	33.9	36.2	40.6	52.5	55.3	60.4	58.3	53.6	43.7	37.8	34.4		
	0.4%	WB	39.5	40.2	43.2	51.7	61.4	63.1	69.5	65.4	60.5	56.5	45.6	40.5		
		MCDB	41.9	44.5	47.6	64.4	73.0	72.3	83.7	74.0	66.7	60.0	48.3	42.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	37.9	40.8	47.0	58.5	60.4	66.1	62.7	58.1	49.9	43.0	38.2		
		MCDB	38.9	41.2	44.8	56.9	69.5	69.4	77.5	69.6	63.9	54.1	45.7	40.1		
	5%	WB	35.5	36.2	38.7	43.8	55.6	58.0	63.5	60.8	56.1	47.1	40.6	36.2		
		MCDB	37.4	39.1	42.7	50.8	66.3	66.1	72.5	67.7	61.6	51.1	43.1	38.3		
	10%	WB	33.5	34.2	36.7	41.2	52.6	55.8	61.2	58.8	53.9	44.3	38.1	34.5		
		MCDB	35.3	37.1	40.1	47.0	61.9	62.6	69.2	65.1	58.7	47.4	40.5	36.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.9	14.8	16.9	18.3	16.9	17.9	17.7	16.6	13.1	10.3	10.5		
		MCDBR	9.8	12.3	15.4	23.8	28.9	24.8	26.9	23.6	20.9	16.4	11.9	11.0		
		MCWBR	7.8	8.9	10.2	13.6	15.4	13.7	14.0	13.2	13.1	11.5	9.1	9.0		
		MCDBR	9.6	12.3	14.0	20.8	27.3	23.2	24.9	21.0	20.7	15.7	12.2	10.9		
	5% WB	MCWBR	7.8	9.1	9.8	12.6	15.3	13.3	13.5	12.4	13.3	11.5	9.7	9.2		
		MCWBR	7.8	9.1	9.8	12.6	15.3	13.3	13.5	12.4	13.3	11.5	9.7	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.209	0.268	0.300	0.329	0.338	0.339	0.350	0.337	0.310	0.290	0.223	N/A		
	taud		2.041	2.219	2.229	2.288	2.407	2.463	2.457	2.486	2.546	2.455	2.118	N/A		
	Ebn at Noon		141	217	253	271	279	280	273	267	255	212	133	N/A		
	Edh at Noon		10	19	28	33	32	31	30	27	21	16	9	N/A		
	RadAvg		61	247	684	1263	1532	1555	1467	1131	696	309	88	30		
(45) All-Sky Solar Radiation	RadStd		8	34	76	87	92	138	145	115	70	39	14	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-257	-340	+121	N/A

Nomenclature: See separate page